

UC SAN DIEGO / ECE 251C

5

Gibbs phenomena and the need for localization

Fourier series, low-pass projections, and local representations

Why do more Fourier modes sharpen a jump without removing the overshoot?

5.1 Fourier analysis on a finite interval

We now observe a signal on $[0, 1]$, rather than on all of $\mathbb{R}$.

$$L^2[0, 1] := \left\{ f : [0, 1] \rightarrow \mathbb{C} : \int_0^1 |f(t)|^2 dt < \infty \right\}.$$

The inner product induces the norm:

$$\langle f, g \rangle_{L^2} = \int_0^1 f(t)g^*(t) dt, \quad \|f\|_{L^2} = \sqrt{\langle f, f \rangle_{L^2}}. \quad (5.1)$$

Recall. Conjugate the *second* function: $f(t)f^*(t) = |f(t)|^2$. The inner product of a signal with itself gives its energy. Endpoint values do not affect these integrals.

Period-one exponentials. For $e^{j\omega t}$ to repeat after one unit, we need

$$e^{j\omega(t+1)} = e^{j\omega t} \iff e^{j\omega} = 1 \iff \omega = 2\pi k, \quad k \in \mathbb{Z}.$$

Define $e_k(t) := e^{j2\pi kt}$. Then $e_k(t+1) = e_k(t)$. Here 1 is a period, not necessarily the *smallest* period; $e_0 = 1$ is constant.

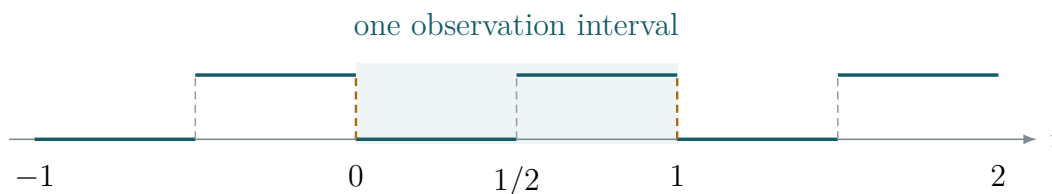


Figure 5.1: A step on $[0, 1]$, repeated periodically. The endpoint mismatch creates a second jump where consecutive periods meet.

No endpoint-matching condition is required for membership in $L^2[0, 1]$. Periodic extension is how the Fourier representation repeats the observed signal.

5.2 Fourier coefficients are inner products

Definition. Two functions are *orthogonal* if their inner product is zero. A family is *orthonormal* if distinct members are orthogonal and every member has norm one.

The Fourier functions $\{e_k\}_{k \in \mathbb{Z}}$ are orthonormal. For integers k, ℓ , we calculate

$$\begin{aligned} \langle e_k, e_\ell \rangle_{L^2} &= \int_0^1 e^{j2\pi kt} e^{-j2\pi \ell t} dt = \int_0^1 e^{j2\pi(k-\ell)t} dt \\ &= \begin{cases} 1, & k = \ell, \\ \left[\frac{e^{j2\pi(k-\ell)t}}{j2\pi(k-\ell)} \right]_0^1 = 0, & k \neq \ell. \end{cases} \end{aligned}$$

Thus $\langle e_k, e_\ell \rangle_{L^2} = \delta[k - \ell]$. In particular, $\|e_k\|_{L^2} = 1$.

Recall. For a T -periodic signal, the Fourier frequencies are $k\omega_0$, where $\omega_0 = 2\pi/T$, and

$$c_k = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) e^{-jk\omega_0 t} dt.$$

Here $T = 1$. This gives the following definitions.

Definition. The Fourier coefficients and the N th partial sum are

$$c_k := \langle f, e_k \rangle_{L^2} = \int_0^1 f(t) e^{-j2\pi kt} dt, \quad (5.2)$$

$$S_N f := \sum_{k=-N}^N c_k e_k, \quad N = 0, 1, 2, \dots \quad (5.3)$$

Analysis uses e_k^* ; reconstruction uses e_k .

The coefficients are well defined. The elementary inequality

$$(|f(t)| - 1)^2 \geq 0 \implies 2|f(t)| \leq 1 + |f(t)|^2$$

gives

$$2 \int_0^1 |f(t)| dt \leq 1 + \int_0^1 |f(t)|^2 dt < \infty.$$

Since $|e_k| = 1$, the absolute value of the integrand in [equation \(5.2\)](#) is $|f|$. Every coefficient is therefore well defined.

Extracting the coordinates. To see why the inner products give the coefficients, suppose first that $f = \sum_{k=-N}^N a_k e_k$. For $|\ell| \leq N$,

$$\langle f, e_\ell \rangle_{L^2} = \sum_{k=-N}^N a_k \langle e_k, e_\ell \rangle_{L^2} = \sum_{k=-N}^N a_k \delta[k - \ell] = a_\ell.$$

All cross terms vanish. The inner product extracts the coordinate of f along the chosen Fourier function.

Example. Find the coefficients of

$$f(t) = 3 + 2 \cos(2\pi t) - \sin(4\pi t).$$

Using $2 \cos \theta = e^{j\theta} + e^{-j\theta}$ and $\sin \theta = (e^{j\theta} - e^{-j\theta})/(2j)$,

$$\begin{aligned} f(t) &= 3 + e^{j2\pi t} + e^{-j2\pi t} - \frac{e^{j4\pi t} - e^{-j4\pi t}}{2j} \\ &= 3e_0(t) + e_1(t) + e_{-1}(t) + \frac{j}{2}e_2(t) - \frac{j}{2}e_{-2}(t). \end{aligned}$$

Here $1/j = -j$. Reading off the coordinates gives

$$c_0 = 3, \quad c_1 = c_{-1} = 1, \quad c_2 = \frac{j}{2}, \quad c_{-2} = -\frac{j}{2};$$

all other coefficients are zero.

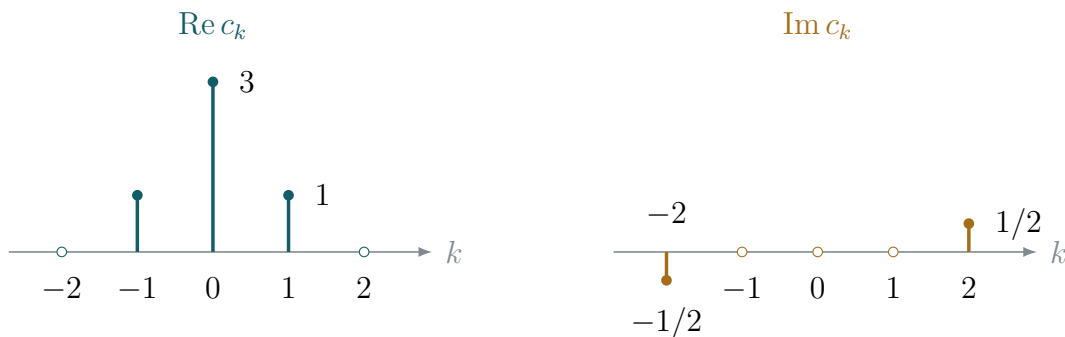


Figure 5.2: The real part of this spectrum is even; the imaginary part is odd. The coefficient at $k = -2$ is the conjugate of the coefficient at $k = 2$.

For any real-valued f , conjugating the coefficient integral gives

$$c_k^* = \int_0^1 f(t) e^{j2\pi kt} dt = c_{-k}.$$

Consequently $c_k e_k + c_{-k} e_{-k} = 2 \operatorname{Re}(c_k e_k)$ is real. Complex coordinates do not make the reconstructed real signal complex.

5.3 The best approximation with low frequencies

Let $V_N := \text{span}\{e_{-N}, \dots, e_N\}$ denote the space of trigonometric polynomials $p = \sum_{k=-N}^N a_k e_k$. Its angular frequencies have magnitude at most $2\pi N$.

Proposition 5.1. $S_N f$ is the unique minimizer of $\|f - p\|_{L^2}$ over $p \in V_N$.

Proof. Set $r = f - S_N f$. For every $|\ell| \leq N$,

$$\langle r, e_\ell \rangle_{L^2} = c_\ell - \sum_{k=-N}^N c_k \langle e_k, e_\ell \rangle_{L^2} = c_\ell - c_\ell = 0.$$

Thus r is orthogonal to every element of V_N . For any $p \in V_N$, write $f - p = r + (S_N f - p)$ and expand:

$$\begin{aligned} \|f - p\|_{L^2}^2 &= \|r\|_{L^2}^2 + \|S_N f - p\|_{L^2}^2 + 2 \operatorname{Re} \langle r, S_N f - p \rangle_{L^2} \\ &= \|f - S_N f\|_{L^2}^2 + \|S_N f - p\|_{L^2}^2. \end{aligned}$$

The last term is nonnegative and vanishes exactly when $p = S_N f$ in L^2 . $\square$

This makes S_N an *orthogonal projector*. It is also a low-pass filter in Fourier-series coordinates:

$$S_N e_k = \begin{cases} e_k, & |k| \leq N, \\ 0, & |k| > N. \end{cases}$$

It keeps the low frequencies with gain one and removes the rest.

Theorem 5.2 (Fourier series). The family $\{e_k\}_{k \in \mathbb{Z}}$ is a complete orthonormal basis of $L^2[0, 1]$. In particular,

$$\boxed{f = \sum_{k \in \mathbb{Z}} \langle f, e_k \rangle_{L^2} e_k \quad \text{in } L^2, \quad \|f\|_{L^2}^2 = \sum_{k \in \mathbb{Z}} |c_k|^2.} \quad (5.4)$$

The first identity means $\|f - S_N f\|_{L^2} \rightarrow 0$; the second is Parseval's identity. Completeness is a standard Fourier-series result and does not come from orthogonality alone.

For a finite partial sum, orthogonality gives

$$\|S_N f\|_{L^2}^2 = \sum_{k=-N}^N |c_k|^2, \quad \|f - S_N f\|_{L^2}^2 = \|f\|_{L^2}^2 - \sum_{k=-N}^N |c_k|^2. \quad (5.5)$$

Indeed, expand the first squared norm and cancel the cross terms; then use $f = (f - S_N f) + S_N f$, an orthogonal decomposition. Completeness makes the last error tend to zero and yields the infinite energy identity in [equation \(5.4\)](#).

5.4 Fourier truncation as periodic convolution

Insert the coefficient integrals into the finite sum:

$$\begin{aligned} (S_N f)(t) &= \sum_{k=-N}^N \left(\int_0^1 f(u) e^{-j2\pi k u} du \right) e^{j2\pi k t} \\ &= \int_0^1 f(u) \underbrace{\sum_{k=-N}^N e^{j2\pi k(t-u)} du}_{D_N(t-u)} = (f *_{\text{per}} D_N)(t). \end{aligned} \quad (5.6)$$

Definition. $D_N(s) := \sum_{k=-N}^N e^{j2\pi k s}$ is the *Dirichlet kernel*. The integral in equation (5.6) is *periodic convolution*: arguments outside $[0, 1]$ wrap around by periodicity. Unlike convolution on $\mathbb{R}$, we integrate over one period.

Calculate the kernel. For $s \notin \mathbb{Z}$, let $q = e^{j2\pi s} \neq 1$. A geometric sum gives

$$\begin{aligned} D_N(s) &= q^{-N} \sum_{r=0}^{2N} q^r = e^{-j2\pi N s} \frac{1 - e^{j2\pi(2N+1)s}}{1 - e^{j2\pi s}} \\ &= e^{-j2\pi N s} \frac{-2j e^{j\pi(2N+1)s} \sin((2N+1)\pi s)}{-2j e^{j\pi s} \sin(\pi s)} = \frac{\sin((2N+1)\pi s)}{\sin(\pi s)}. \end{aligned}$$

All phase factors cancel. If $s \in \mathbb{Z}$, use the original sum: $D_N(s) = 2N + 1$.

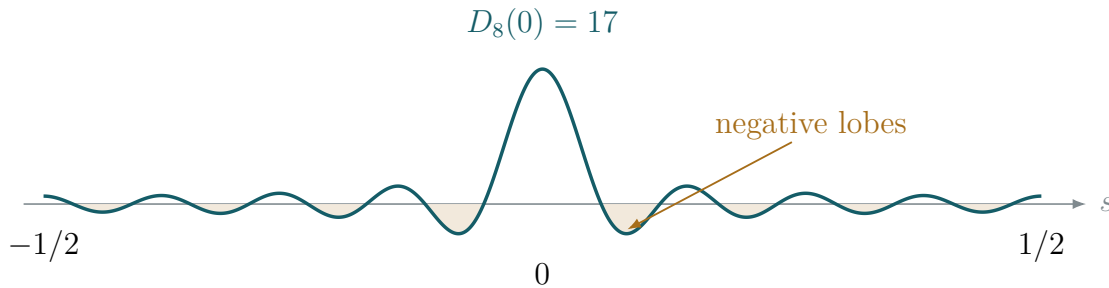


Figure 5.3: The Dirichlet kernel for $N = 8$. Shading marks the negative lobes. Its signed area over one period is one, but the kernel is not nonnegative.

Two checks. Each exponential is 1-periodic, so $D_N(s + 1) = D_N(s)$. Also,

$$\int_0^1 D_N(s) ds = \sum_{k=-N}^N \int_0^1 e^{j2\pi k s} ds = 1.$$

Only $k = 0$ contributes. The kernel is real, since pairing k with $-k$ gives $D_N(s) = 1 + 2 \sum_{k=1}^N \cos(2\pi k s)$.

Unit integral does not make this a positive average. For $N \geq 1$, the kernel has negative lobes. Even when $0 \leq f \leq 1$, the output $S_N f$ need not stay between zero and one.

5.5 Fourier coefficients of a step

Example. Let $f(t) = \mathbf{1}_{[1/2, 1)}(t)$ on $[0, 1]$. Then

$$c_0 = \int_{1/2}^1 1 \, dt = \frac{1}{2}.$$

For $k \neq 0$,

$$\begin{aligned} c_k &= \int_{1/2}^1 e^{-j2\pi kt} \, dt = \left[\frac{e^{-j2\pi kt}}{-j2\pi k} \right]_{1/2}^1 \\ &= \frac{(-1)^k - 1}{j2\pi k} = \begin{cases} 0, & k \neq 0 \text{ even,} \\ \frac{j}{\pi k}, & k \text{ odd.} \end{cases} \end{aligned}$$

No finite Fourier sum equals this step in L^2 . To see this, choose an odd frequency beyond that sum. Its coefficient would be zero by orthogonality, but the coefficient of the step is not zero.

For a positive odd k , combine the positive and negative frequencies:

$$\begin{aligned} c_k e_k(t) + c_{-k} e_{-k}(t) &= \frac{j}{\pi k} (e^{j2\pi kt} - e^{-j2\pi kt}) \\ &= -\frac{2}{\pi k} \sin(2\pi kt). \end{aligned}$$

Set $k = 2r - 1$, for $r = 1, \dots, M$. Since the even coefficients vanish,

$$s_M(t) := (S_{2M-1}f)(t) = (S_{2M}f)(t) = \frac{1}{2} - \frac{2}{\pi} \sum_{r=1}^M \frac{\sin(2\pi(2r-1)t)}{2r-1}. \quad (5.7)$$

There are $M + 1$ nonzero *real* coefficients: the mean and M sine amplitudes. The highest angular frequency is $2\pi(2M - 1)$.

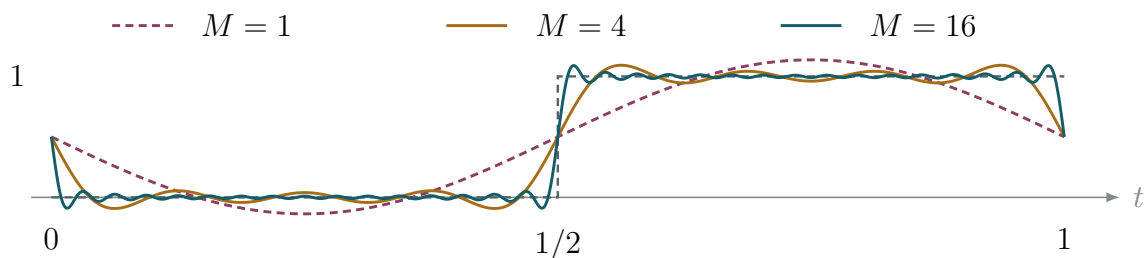


Figure 5.4: The same step with one, four, and sixteen positive odd frequencies. More frequencies sharpen the transition, but ringing remains on both sides of the jump.

5.6 Gibbs overshoot does not go away

Calculate the location. Write $t = 1/2 + x$. Since $2r - 1$ is odd,

$$\sin(2\pi(2r - 1)(1/2 + x)) = -\sin(2\pi(2r - 1)x).$$

Thus

$$s_M(1/2 + x) = \frac{1}{2} + \frac{2}{\pi} \sum_{r=1}^M \frac{\sin(2\pi(2r - 1)x)}{2r - 1}. \quad (5.8)$$

In particular, $s_M(1/2) = 1/2$ for every M . Differentiate the *finite* sum:

$$\frac{d}{dx} s_M(1/2 + x) = 4 \sum_{r=1}^M \cos(2\pi(2r - 1)x).$$

To add the cosines, use

$$2 \sin \alpha \cos((2r - 1)\alpha) = \sin(2r\alpha) - \sin((2r - 2)\alpha).$$

The sum telescopes:

$$\begin{aligned} 2 \sin \alpha \sum_{r=1}^M \cos((2r - 1)\alpha) &= [\sin(2\alpha) - \sin 0] + [\sin(4\alpha) - \sin(2\alpha)] \\ &\quad + \cdots + [\sin(2M\alpha) - \sin((2M - 2)\alpha)] \\ &= \sin(2M\alpha). \end{aligned}$$

Put $\alpha = 2\pi x$. For $0 < x < 1/2$,

$$\frac{d}{dx} s_M(1/2 + x) = 2 \frac{\sin(4\pi Mx)}{\sin(2\pi x)}. \quad (5.9)$$

The denominator is positive. The numerator first changes from positive to negative when $4\pi Mx = \pi$. Therefore

$$\boxed{t_M = \frac{1}{2} + \frac{1}{4M}}. \quad (5.10)$$

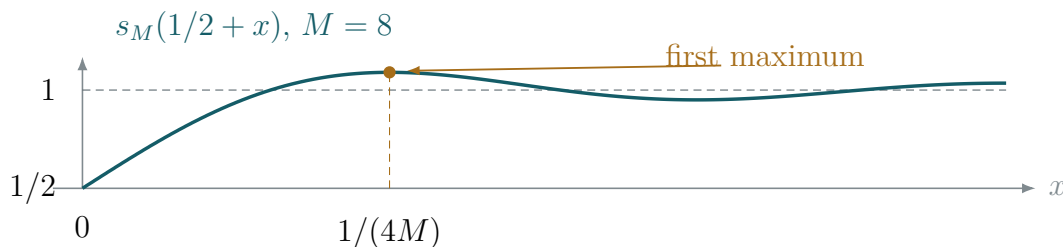


Figure 5.5: Zoom on the first peak: the vertical scale starts at $1/2$. The maximum is exactly $1/(4M)$ to the right of the jump.

Calculate the height. Substitute t_M into equation (5.8):

$$s_M(t_M) = \frac{1}{2} + \frac{2}{\pi} \sum_{r=1}^M \frac{\sin\left(\pi \frac{2r-1}{2M}\right)}{2r-1}.$$

Set

$$q_r := \frac{r-1/2}{M} = \frac{2r-1}{2M}, \quad 2r-1 = 2Mq_r.$$

These are the midpoints of the M equal subintervals of $[0, 1]$. Hence

$$s_M(t_M) = \frac{1}{2} + \frac{1}{\pi} \underbrace{\frac{1}{M} \sum_{r=1}^M \frac{\sin(\pi q_r)}{q_r}}_{\text{midpoint Riemann sum on } [0,1]}. \quad (5.11)$$

The function $q \mapsto \sin(\pi q)/q$ extends continuously to $q = 0$ with value π , so the Riemann sum tends to its integral.

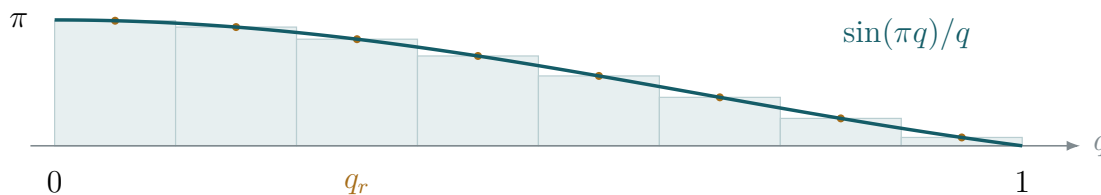


Figure 5.6: The peak height is determined by a midpoint Riemann sum. The apparent singularity at $q = 0$ is removable: the curve starts at height π .

With $v = \pi q$, we have $dq = dv/\pi$. Thus

$$\int_0^1 \frac{\sin(\pi q)}{q} dq = \int_0^\pi \frac{\sin v}{v/\pi} \frac{dv}{\pi} = \int_0^\pi \frac{\sin v}{v} dv.$$

Consequently,

$$\lim_{M \rightarrow \infty} (s_M(t_M) - 1) = \frac{1}{\pi} \int_0^\pi \frac{\sin v}{v} dv - \frac{1}{2} \approx 0.089489872. \quad (5.12)$$

The first peak tends to $1.08949\dots$, not to 1. The persistent overshoot is about 8.95% of this unit jump.

M	distance $t_M - 1/2$	peak $s_M(t_M)$	overshoot
1	$1/4$	1.136620	13.66%
4	$1/16$	1.092113	9.21%
16	$1/64$	1.089653	8.97%

5.7 Small energy is not small error everywhere

The persistent Gibbs overshoot does not contradict convergence in L^2 . The L^2 error is an *integral*, not an error bound at every point. Here $\|f\|_{L^2}^2 = 1/2$. The mean contributes $1/4$ of the energy; each pair of odd Fourier coefficients contributes $2/(\pi^2(2r-1)^2)$. By equation (5.5),

$$\|f - s_M\|_{L^2}^2 = \frac{1}{4} - \frac{2}{\pi^2} \sum_{r=1}^M \frac{1}{(2r-1)^2}. \quad (5.13)$$

Parseval applied to the full step gives

$$\frac{1}{2} = \frac{1}{4} + \frac{2}{\pi^2} \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}.$$

Subtracting the retained energy therefore gives

$$\|f - s_M\|_{L^2}^2 = \frac{2}{\pi^2} \sum_{r=M+1}^{\infty} \frac{1}{(2r-1)^2} \rightarrow 0. \quad (5.14)$$

For example, the tail is bounded by $\sum_{r>M} r^{-2} \leq \int_M^{\infty} x^{-2} dx = 1/M$. Its disappearance is compatible with the nonzero peak error at the moving point t_M .

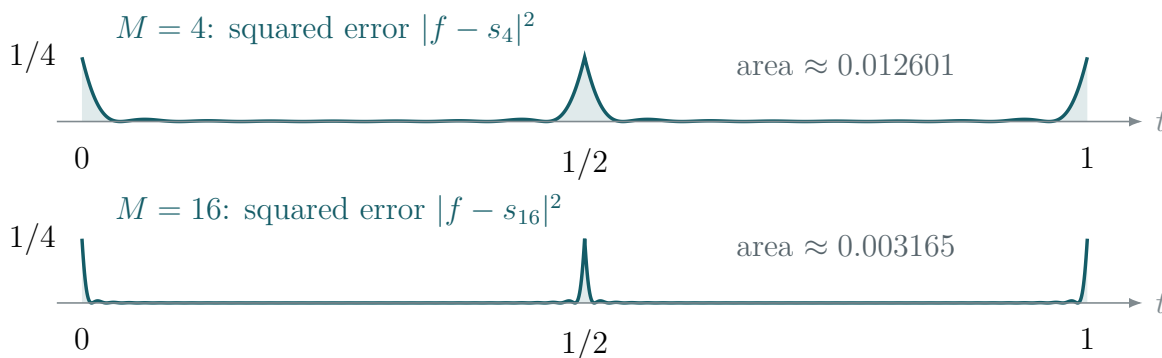


Figure 5.7: The plotted quantity is squared error; its shaded area is the L^2 error squared. The large values occupy narrower regions as M increases. The spikes at the interval endpoints reflect the periodic jump there.

Gibbs phenomenon

The ringing becomes narrower, but not arbitrarily shorter. Its first peak approaches the jump at distance $1/(4M)$, while its excess height tends to $0.08949\dots$. The total squared error nevertheless tends to zero.

5.8 Fejér averaging

Overshoot is not unavoidable in a Fourier representation. We can remove it by changing the weights without changing the basis.

Fejér averaging. Average the first $N + 1$ partial sums:

$$\begin{aligned} A_N f &:= \frac{1}{N+1} \sum_{m=0}^N S_m f \\ &= \frac{1}{N+1} \sum_{m=0}^N \sum_{|k| \leq m} c_k e_k = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1}\right) c_k e_k. \end{aligned} \quad (5.15)$$

The coefficient c_k occurs when $m = |k|, \dots, N$: exactly $N + 1 - |k|$ times. This explains the triangular weights.

Just as in [equation \(5.6\)](#), $A_N f = f *_{\text{per}} F_N$. The kernel is

$$\begin{aligned} F_N(s) &= \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1}\right) e^{j2\pi ks} \\ &= \frac{1}{N+1} \left| \sum_{r=0}^N e^{j2\pi rs} \right|^2 \geq 0. \end{aligned} \quad (5.16)$$

To verify the second equality, expand the square into $\sum_{r=0}^N \sum_{\ell=0}^N e^{j2\pi(r-\ell)s}$. There are $N + 1 - |k|$ pairs with $r - \ell = k$. Integrating leaves only the $N + 1$ pairs with $r = \ell$, so $\int_0^1 F_N = 1$.

For a real signal with $0 \leq f \leq 1$, positivity and unit integral imply

$$0 \leq (A_N f)(t) = \int_0^1 f(u) F_N(t - u) du \leq \int_0^1 F_N(t - u) du = 1.$$

Thus Fejér averaging cannot overshoot the range of the step.

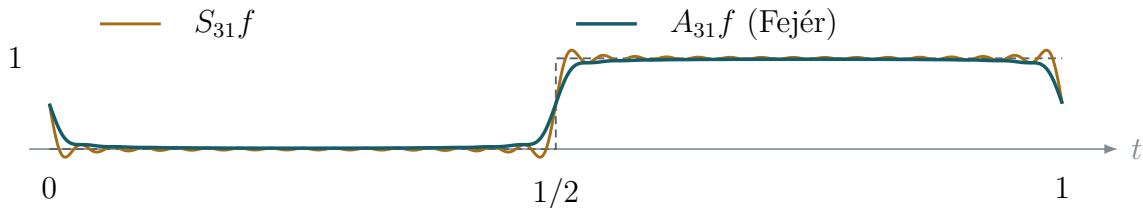


Figure 5.8: Same Fourier frequencies, different weights. Fejér averaging stays within $[0, 1]$, but rounds the transition. The sharp projector is the least-squares approximation; the averaged operator is not.

The averaged operator A_N is not the projector S_N : nonzero retained frequencies have gain below one. The Fourier building blocks are still global.

5.9 Localize the representation

To change part of an approximation without changing it everywhere, use building blocks supported on small intervals. In contrast, each Fourier function has $|e_k(t)| = 1$ throughout $[0, 1]$. Changing one Fourier coefficient changes the approximation throughout that interval.

For $j = 0, 1, \dots$ and $k = 0, \dots, 2^j - 1$, define the dyadic cells

$$I_{j,k} := [k2^{-j}, (k+1)2^{-j}), \quad |I_{j,k}| = 2^{-j}. \quad (5.17)$$

There are 2^j cells. The index j sets their size; k identifies a location. Endpoints can be assigned either value without changing an L^2 function.

Definition. The normalized cell functions are

$$\phi_{j,k}(t) := 2^{j/2} \mathbf{1}_{I_{j,k}}(t) = \begin{cases} 2^{j/2}, & t \in I_{j,k}, \\ 0, & t \notin I_{j,k}. \end{cases} \quad (5.18)$$

The height compensates for the width:

$$\|\phi_{j,k}\|_{L^2}^2 = \int_{I_{j,k}} |2^{j/2}|^2 dt = 2^j |I_{j,k}| = 2^j 2^{-j} = 1.$$

At a *fixed* scale j , two different cells are disjoint, so

$$\langle \phi_{j,k}, \phi_{j,\ell} \rangle_{L^2} = 0 \quad (k \neq \ell).$$

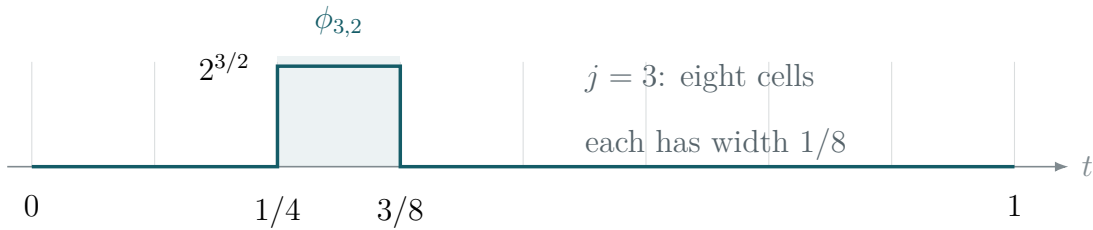


Figure 5.9: At a fixed scale, the normalized indicators have disjoint nonzero regions. Changing the coefficient of $\phi_{3,2}$ changes only the highlighted cell.

If g has value b_k on each cell $I_{j,k}$, then

$$g = \sum_{k=0}^{2^j-1} b_k \mathbf{1}_{I_{j,k}} = \sum_{k=0}^{2^j-1} (2^{-j/2} b_k) \phi_{j,k}.$$

Thus, at each fixed scale j , these functions form an orthonormal basis for the cellwise-constant signals.

Calculate the local projection. As in Section 5.3, project onto the span of the orthonormal family:

$$P_j f = \sum_{k=0}^{2^j-1} \langle f, \phi_{j,k} \rangle_{L^2} \phi_{j,k}. \quad (5.19)$$

Since the cell functions are real,

$$\langle f, \phi_{j,k} \rangle_{L^2} = \int_0^1 f(u) \phi_{j,k}(u) du = 2^{j/2} \int_{I_{j,k}} f(u) du.$$

For $t \in I_{j,k}$, all terms except the k th vanish. Therefore

$$\begin{aligned} (P_j f)(t) &= \left(2^{j/2} \int_{I_{j,k}} f(u) du \right) 2^{j/2} \\ &= 2^j \int_{I_{j,k}} f(u) du = \boxed{\frac{1}{|I_{j,k}|} \int_{I_{j,k}} f(u) du}. \end{aligned} \quad (5.20)$$

The least-squares approximation is the *average on each cell*. This is the local analogue of retaining Fourier coordinates.

Example. Move the jump to $1/3$: let $f_{1/3} = \mathbf{1}_{[1/3,1]}$. At $j = 3$, the jump lies in $I_{3,2} = [1/4, 3/8]$. The value in this cell is

$$8 \int_{1/4}^{3/8} f_{1/3}(u) du = 8 \left(\frac{3}{8} - \frac{1}{3} \right) = \frac{1}{3}.$$

All cells to its left have average zero; all cells to its right have average one.

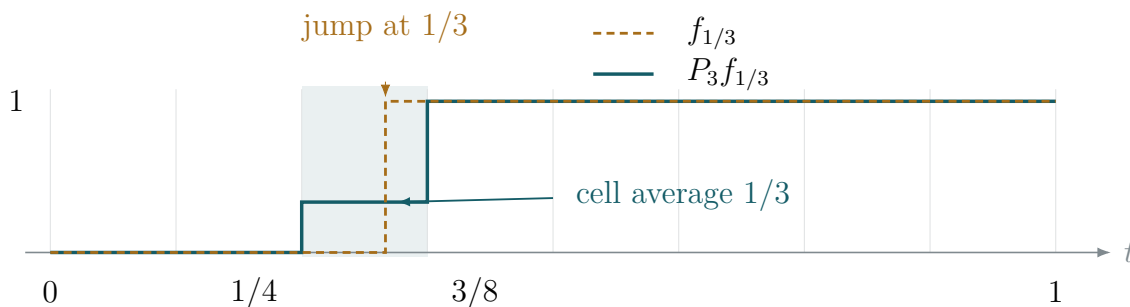


Figure 5.10: Local averaging replaces the portion of the step inside $[1/4, 3/8]$ by its mean. All the approximation error is confined to that one cell.

Only the cell containing the jump has an approximation error. No oscillatory correction is spread across the interval. If $0 \leq f \leq 1$, every cell average also lies between zero and one.

The midpoint step from Section 5.5 is even simpler: it is already constant on the two cells at $j = 1$, so $P_1 f = f$. This suggests replacing global sinusoids by local averages and local differences.

5.10 From local averages to wavelets

Definition. The Haar *scaling function* and *wavelet* are

$$\phi(t) := \mathbf{1}_{[0,1)}(t), \quad \psi(t) := \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).$$

They describe a background level and a difference between two halves.

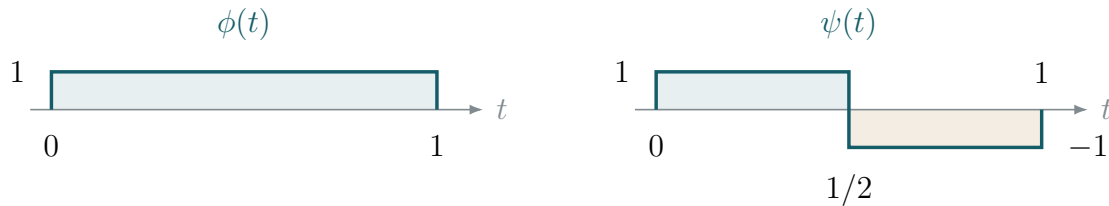


Figure 5.11: An average and a contrast. The wavelet has equal positive and negative areas. Vertical segments mark jumps, not extra values of the function.

Since $\int_0^1 \psi(t) dt = 1/2 - 1/2 = 0$, the positive and negative areas cancel. Both functions have energy one:

$$\|\phi\|_{L^2}^2 = 1, \quad \|\psi\|_{L^2}^2 = \frac{1}{2} + \frac{1}{2} = 1, \quad \langle \phi, \psi \rangle_{L^2} = \int_0^1 \psi(t) dt = 0.$$

Thus ϕ, ψ are an orthonormal pair.

Example. Return to $f = \mathbf{1}_{[1/2,1)}$. Its two coefficients are

$$\langle f, \phi \rangle_{L^2} = \int_{1/2}^1 1 dt = \frac{1}{2}, \quad \langle f, \psi \rangle_{L^2} = \int_{1/2}^1 (-1) dt = -\frac{1}{2}.$$

$$\boxed{f = \frac{1}{2}\phi - \frac{1}{2}\psi.} \quad (5.21)$$

On the left half this is $1/2 - 1/2 = 0$; on the right it is $1/2 + 1/2 = 1$. Two real coordinates reconstruct the step exactly, without overshoot.

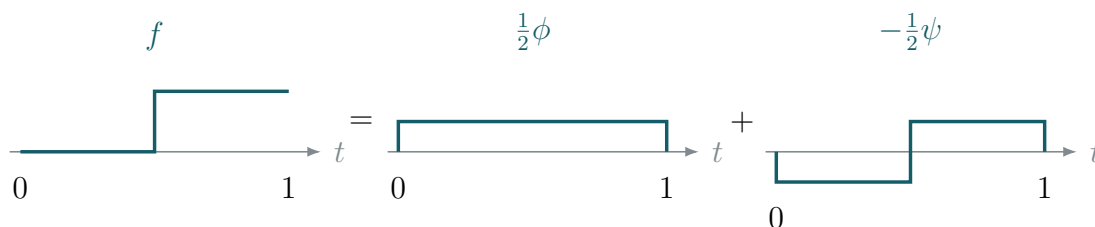


Figure 5.12: The step is a constant background plus a signed local contrast. Unlike a finite trigonometric sum, these two functions reproduce the jump exactly.

This jump aligns with the midpoint. A shifted jump need not have the same two-term representation; locality is the more general idea.

Examine smaller regions. For $j \geq 0$ and $0 \leq k < 2^j$, define

$$\psi_{j,k}(t) := 2^{j/2} \psi(2^j t - k). \quad (5.22)$$

Since $0 \leq 2^j t - k < 1$ exactly when $t \in I_{j,k}$, each wavelet occupies one cell. Increasing j makes it narrower; changing k moves it. Its two halves have opposite signs.

Check the normalization. Put $u = 2^j t - k$, so $dt = 2^{-j} du$. Since ψ vanishes outside $[0, 1]$,

$$\begin{aligned} \|\psi_{j,k}\|_{L^2}^2 &= 2^j 2^{-j} \int_0^1 |\psi(u)|^2 du = 1, \\ \int_0^1 \psi_{j,k}(t) dt &= 2^{j/2} 2^{-j} \int_0^1 \psi(u) du = 0. \end{aligned}$$

Narrower wavelets are taller, but every wavelet has the same energy and zero mean.

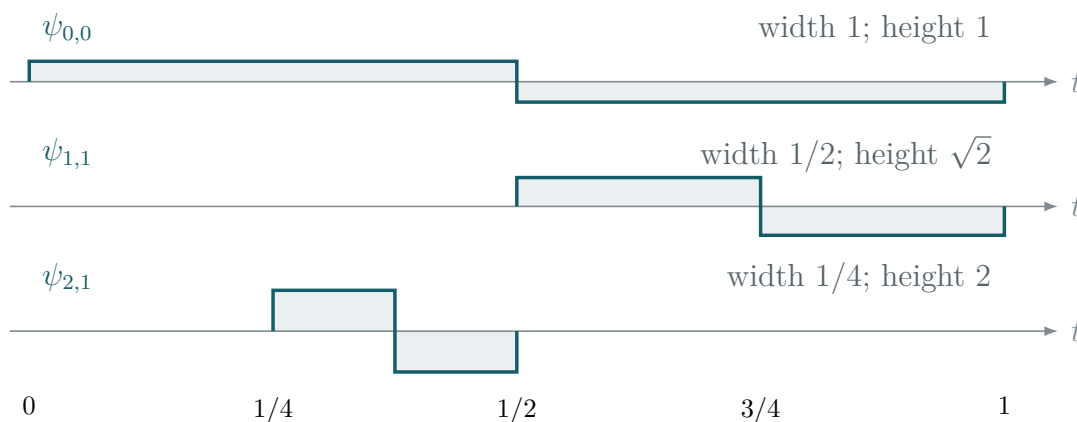


Figure 5.13: The same Haar pattern at different positions and scales, with a common horizontal axis. The amplitude factor keeps the energy equal to one.

Cancellation on constant regions. If $f(t) = b$ on $I_{j,k}$, then

$$\langle f, \psi_{j,k} \rangle_{L^2} = b \int_{I_{j,k}} \psi_{j,k}(t) dt = 0.$$

A constant background cancels; a local contrast need not.

The need for localization

Fourier functions describe global frequency content. Haar wavelets describe local changes at different scales. Their coefficients let us place representational detail where the signal changes, rather than alter the entire interval with each coordinate. This is the starting point for multiresolution signal analysis.